SCHOOL OF BIOTECHNOLOGY

Nile University

CBIO313: Data Mining and Machine Learning

Stroke Prediction Using Machine Learning

Final report

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Abstract

In this project, a machine learning model was developed to predict the risk of stroke using demographic, medical, and lifestyle information. The dataset contains important clinical and personal features such as age, hypertension, heart disease, average glucose level, body mass index, smoking status, work type, and residence type. The target variable was stroke, where 1 represents a stroke case, and 0 represents a non-stroke case.

The project followed a complete machine learning workflow, beginning with data understanding and preprocessing, followed by exploratory data analysis, feature engineering, model training, hyperparameter tuning, evaluation, and deployment. Several classification algorithms were trained and compared, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Gradient Boosting. Because the dataset was highly imbalanced, model performance was not judged by accuracy alone. Instead, recall, precision, F1-score, ROC AUC, confusion matrix, and classification report were used to evaluate the model more fairly.

The final model achieved good recall for detecting stroke cases, meaning it was able to identify most patients who were at risk of stroke. However, the precision for the stroke class was low due to the imbalance in the dataset. Therefore, the model is best interpreted as a screening support tool rather than a final medical diagnostic system. Finally, the trained model was deployed using Streamlit to create an interactive web application that allows users to enter patient information and receive instant stroke-risk prediction.

Introduction

Stroke is a serious medical condition that occurs when blood flow to part of the brain is interrupted or reduced, preventing brain tissue from receiving oxygen and nutrients. Early identification of stroke risk is important because it can help support preventive healthcare decisions and encourage further medical evaluation for high-risk individuals. Machine learning can be useful in this context because it can analyze multiple patient features at the same time and detect patterns that may be associated with stroke risk.

In this project, a stroke prediction dataset was used to build a classification model that predicts whether a patient is likely to be at risk of stroke. The model uses patient information such as age, hypertension, heart disease, glucose level, BMI, smoking status, and other demographic variables. The main objective of the project is to answer the question: Can machine learning predict stroke risk based on clinical, demographic, and lifestyle features?

This project is important because it applies data mining and machine learning techniques to a real-world healthcare problem. However, the model is not intended to replace doctors or clinical diagnosis. Instead, it should be considered a decision-support or screening tool that can help identify patients who may require further medical attention.

Dataset Description

The dataset used in this project is a stroke prediction dataset originally obtained from Kaggle. It contains patient-level information related to demographic, clinical, and lifestyle factors. The original dataset includes 5110 rows and 12 columns before feature engineering. The target column is stroke, where 1 indicates that the patient had a stroke, and 0 indicates that the patient did not have a stroke.

The dataset includes the following types of features:

Demographic features:
Gender, age, marital status, work type, and residence type.

Clinical features:
Hypertension, heart disease, average glucose level, and BMI.

Lifestyle feature:
Smoking status.

Target variable:
Stroke, encoded as 1 for stroke and 0 for no stroke.

The dataset was not fully clean. It contained missing values in the BMI column, unknown values in smoking status, and an ID column that was not useful for prediction. In addition, the target variable was highly imbalanced, with many more non-stroke cases than stroke cases. This imbalance made the project more realistic but also more challenging, because a model could achieve high accuracy simply by predicting most patients as non-stroke.

Methodology

The project followed a complete machine learning lifecycle, beginning with data preprocessing and ending with deployment using Streamlit. The main steps were data cleaning feature engineering, exploratory data analysis, model training, model tuning, evaluation, and deployment.

1. Data Preprocessing

The first step was to load the raw dataset and inspect its structure using pandas. The dataset was checked for shape, data types, missing values, duplicated rows, and target distribution. This step was important because machine learning models require clean and organized data before training.

The ID column was removed because it only identifies individual records and does not provide useful predictive information. Missing BMI values were handled using the median BMI value, which is appropriate because BMI is a numerical variable, and the median is less affected by extreme values than the mean. Unknown values in smoking status were handled during cleaning so that the model could process the column properly.

The dataset was also checked for duplicated rows. No major duplication problem was found. After cleaning, the cleaned dataset was saved as Cleaned_Stroke_Data.csv to make the work reproducible and to allow the cleaned data to be submitted with the project files.

2. Feature Engineering

Feature engineering was performed to create new meaningful variables from existing columns. Three new features were created: age group, BMI category, and glucose risk category. Age group was created by dividing age into categories such as child, young adult, adult, senior, and elderly. The BMI category was created to classify BMI values into underweight, normal, overweight, and obese. Glucose risk was created to classify average glucose levels into normal, prediabetes, and high levels.

These engineered features were useful because they transformed raw numerical values into clinically interpretable categories. This can help the model capture risk patterns more clearly and can also make the Streamlit app output easier to understand.

3. Feature Selection and Preprocessing

The target variable stroke was separated from the input features. The input features were stored in X, and the target variable was stored in y. Numerical and categorical features were processed separately. Numerical features were scaled, while categorical features were encoded using one-hot encoding. This was done using a preprocessing pipeline to ensure that the same transformations were applied consistently during training and prediction.

Feature selection was performed using SelectKBest. This method selects the most relevant features based on their statistical relationship with the target variable. Feature selection helps reduce unnecessary noise, improve model efficiency, and reduce overfitting.

4. Train-Test Split

The dataset was split into training and testing sets using an 80/20 split. Stratification was used during splitting to preserve the same ratio of stroke and non-stroke cases in both the training and testing sets. This was important because the dataset was highly imbalanced. Without stratification, the test set might not properly represent the minority stroke class.

The training set was used to train the models, while the testing set was kept unseen during training and used only for final evaluation. This allowed the project to estimate how well the models could generalize new patient data.

5. Model Training and Comparison

Six machine learning algorithms were trained and compared:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Gradient Boosting

These models were chosen because they represent different types of classification approaches. Logistic Regression is a simple linear model, Decision Tree and Random Forest are tree-based models, SVM can handle complex class boundaries, KNN is a distance-based method, and Gradient Boosting is an ensemble technique that builds models sequentially to improve performance.

The models were evaluated using accuracy, precision, recall, F1-score, and ROC AUC. Since the dataset was highly imbalanced, recall and ROC AUC were considered especially important. Accuracy alone was not enough because a model could achieve high accuracy by mostly predicting the majority class, which is non-stroke.

6. Hyperparameter Tuning

Hyperparameter tuning was performed using GridSearchCV. The goal of tuning was to test different model settings and select the combination that produced the best performance. GridSearchCV also uses cross-validation, which makes the evaluation more reliable because the model is trained and validated on different subsets of the training data.

The tuned final model achieved an accuracy of approximately 0.73, precision of approximately 0.13, recall of 0.80, F1-score of approximately 0.22, and ROC AUC of approximately 0.84. Although the precision was low, the recall was high, which is important in this healthcare screening problem.

Results

The final tuned model showed a good ability to detect stroke-risk cases. The model achieved a recall of 0.80 for the stroke class, meaning that it correctly identified 80% of the actual stroke cases in the test set. This is important because in a medical screening task, missing a high-risk patient can be more serious than incorrectly flagging a low-risk patient.

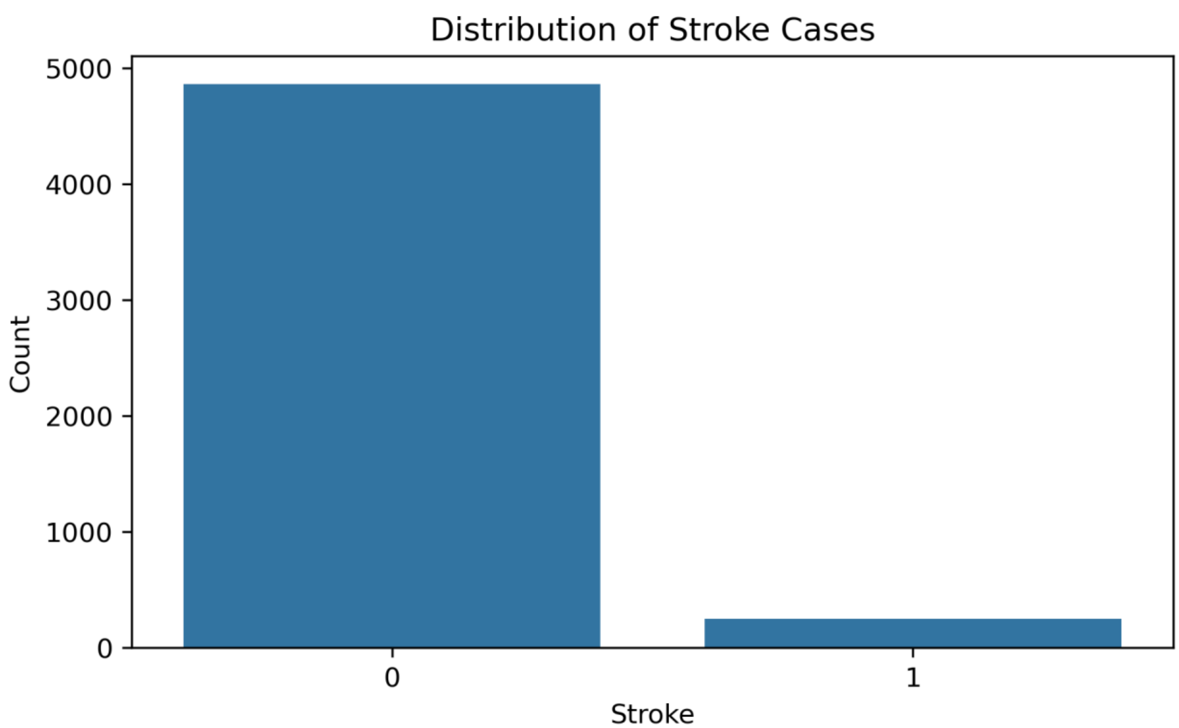
However, the precision for the stroke class was low, around 0.13. This means that many patients predicted as high-risk were actually non-stroke cases. This result is mainly due to the highly imbalanced nature of the dataset, where stroke cases are much fewer than non-stroke cases. Therefore, the model is sensitive to stroke cases but produces false positives.

For the non-stroke class, the model achieved high precision of 0.99, meaning that when the model predicted a patient as non-stroke, it was usually correct. However, the recall for the non-stroke class was lower, around 0.73, because the model classified some non-stroke patients as stroke-risk to avoid missing actual stroke cases.

Overall, the results show that the model is useful as a screening support tool. It can help identify patients who may need further medical evaluation, but it should not be used as a final diagnostic system.

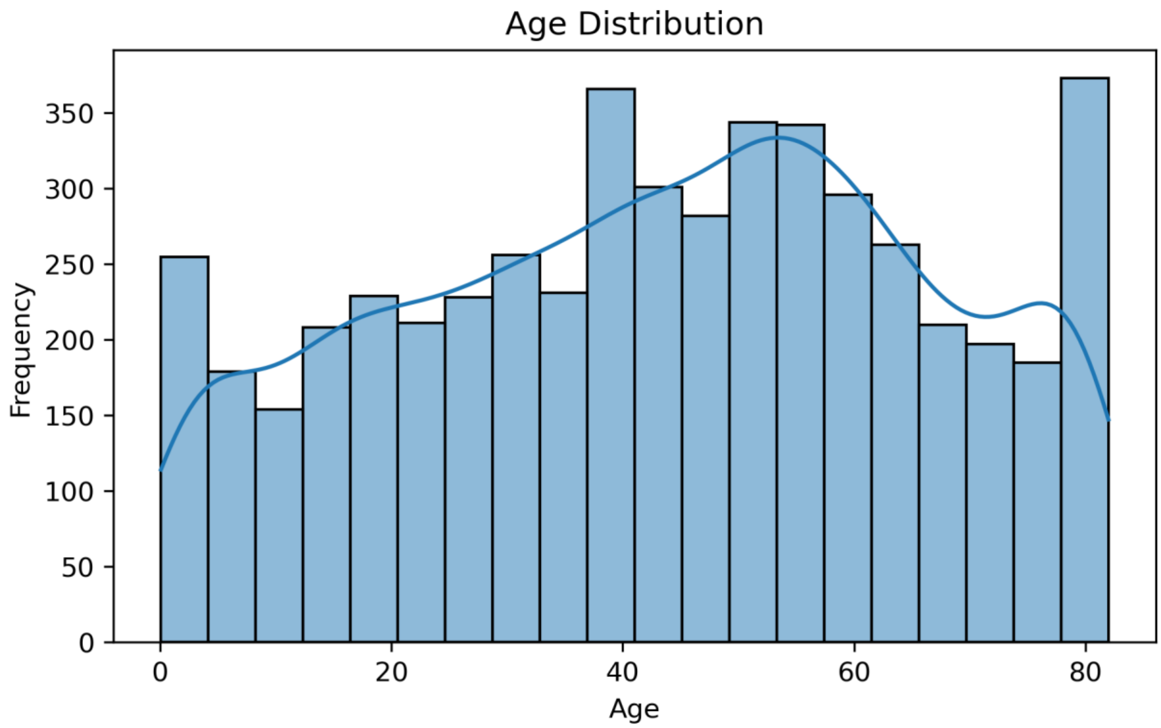
Visualization and Plot Interpretation

Figure 1: Target Distribution



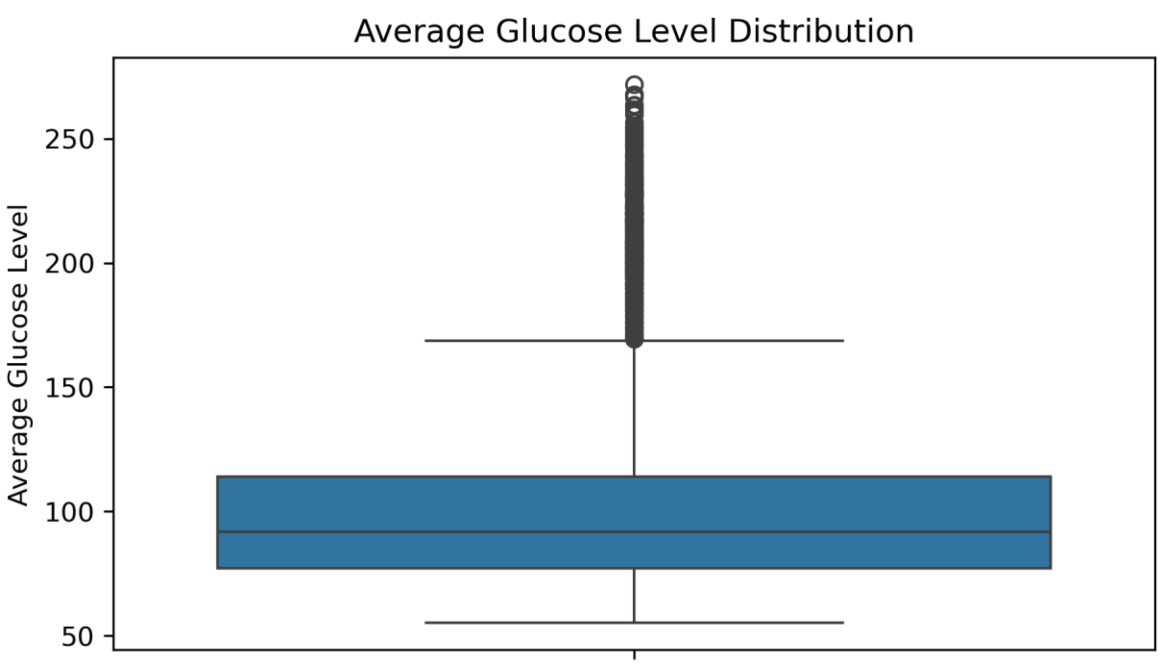
The target distribution plot shows that the dataset is highly imbalanced. Most patients belong to the non-stroke class, while only a small number belong to the stroke class. This is an important finding because it affects how the model should be evaluated. If accuracy alone is used, a model may appear to perform well simply because it predicts the majority of the class most of the time. Therefore, precision, recall, F1-score, ROC AUC, and the confusion matrix were used to provide a more meaningful evaluation.

Figure 2: Age Distribution



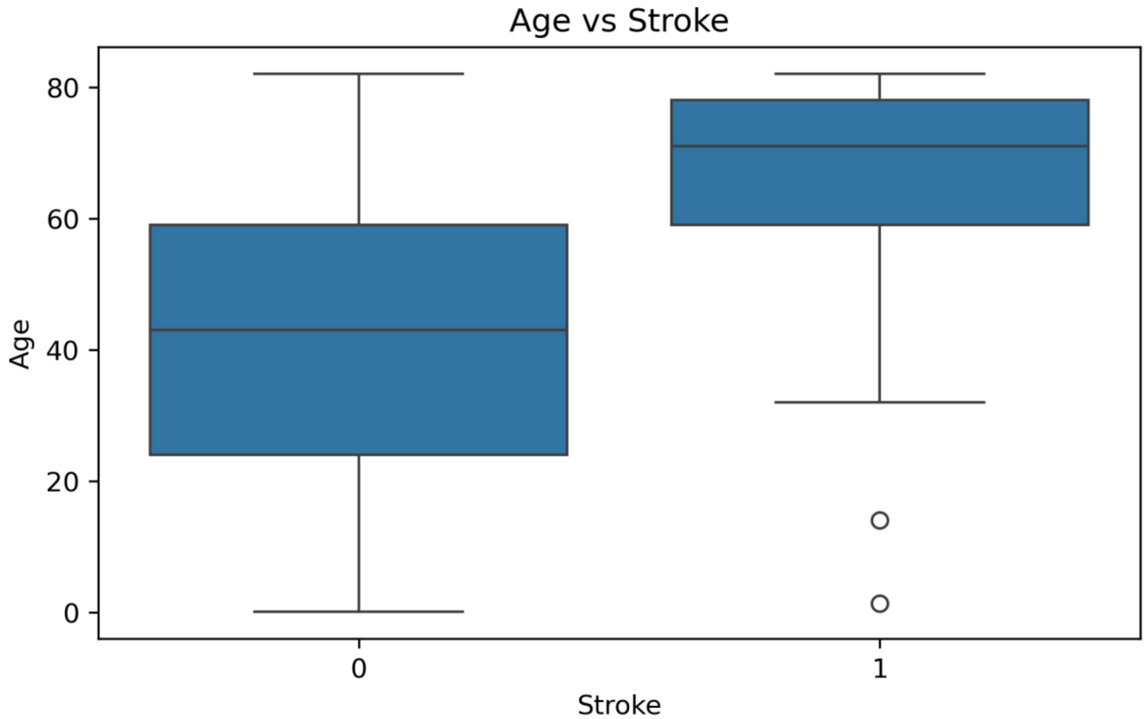
The age distribution plot shows that the dataset includes patients across a wide range of ages. There are both younger and older individuals in the dataset, but stroke risk is expected to be higher among older patients. This plot is useful because it confirms that age is an important feature to investigate in relation to stroke risk. A wide age range also allows the model to learn patterns across different age groups.

Figure 3: Average Glucose Level Distribution



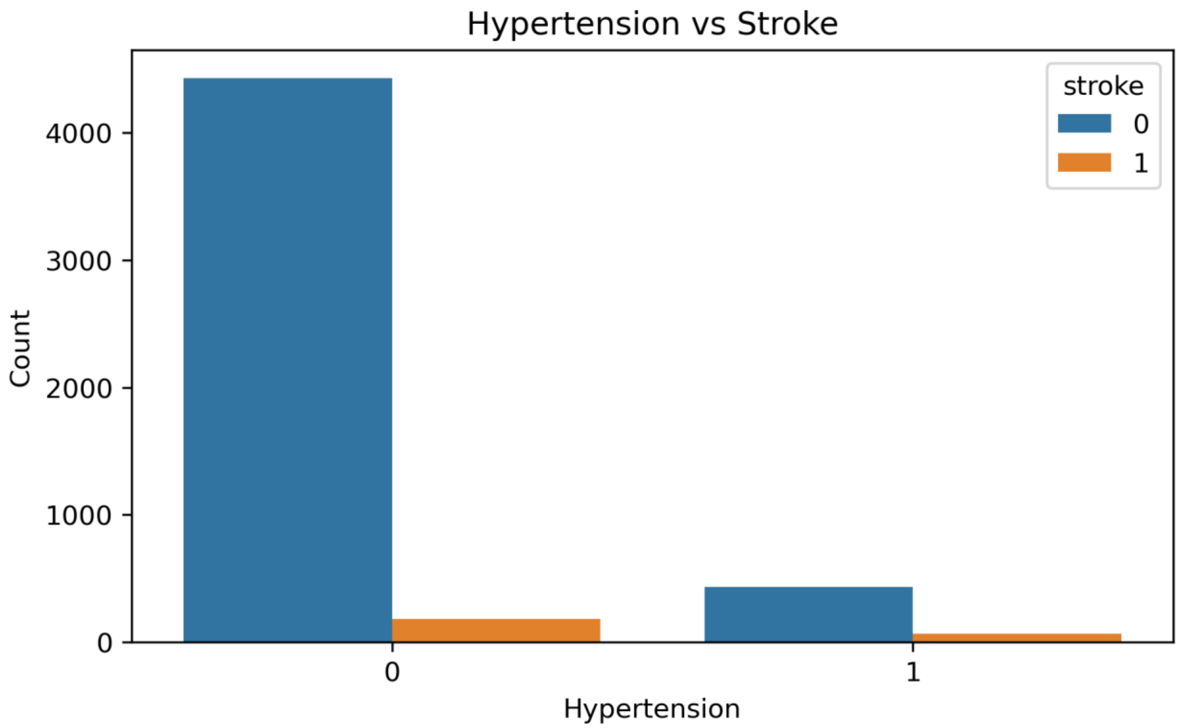
The average glucose level plot shows that most patients have moderate glucose levels, while some patients have very high glucose values. These high values appear as outliers, but they should not automatically be removed because they may represent real high-risk patients. High glucose levels can be clinically meaningful and may contribute to stroke risk, especially when combined with other factors such as age, hypertension, or heart disease.

Figure 4: Age vs Stroke



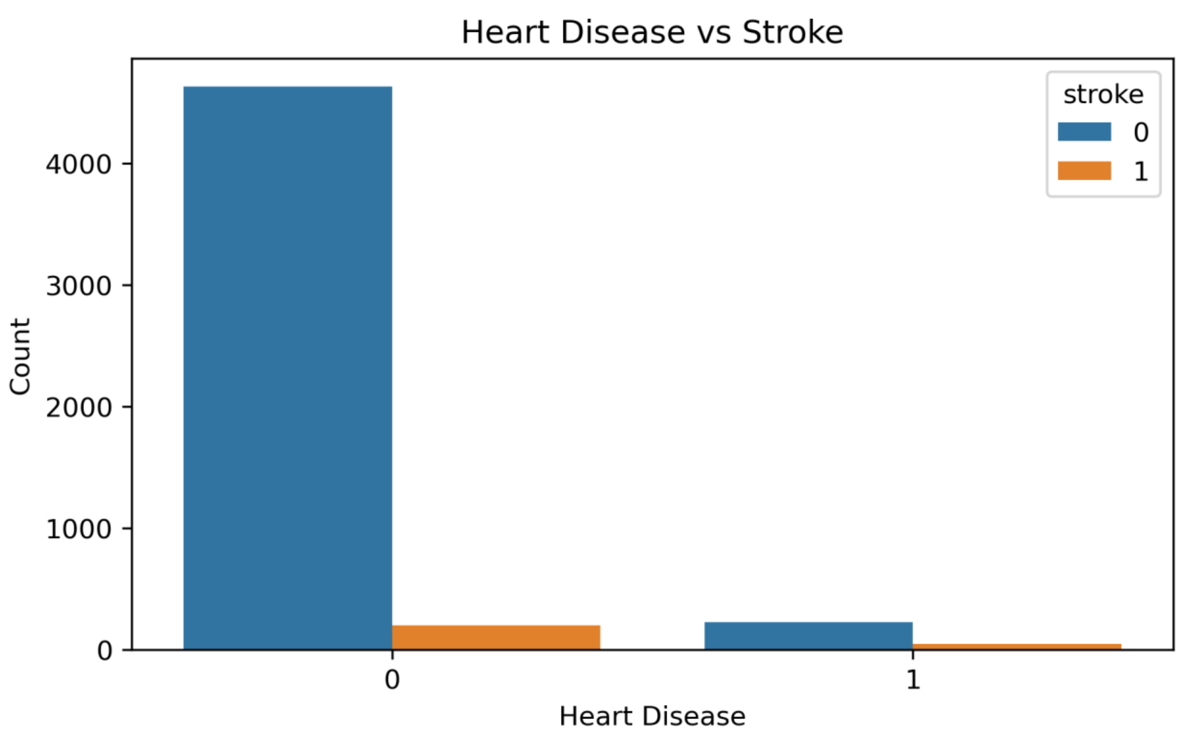
The age versus stroke plot shows that patients who had a stroke were generally older than patients who did not have a stroke. This is one of the strongest visual patterns in the analysis. The median age of stroke patients is higher, suggesting that age is an important predictor of stroke risk. This supports the inclusion of age and the engineered age group feature in the machine learning model.

Figure 5: Hypertension vs Stroke



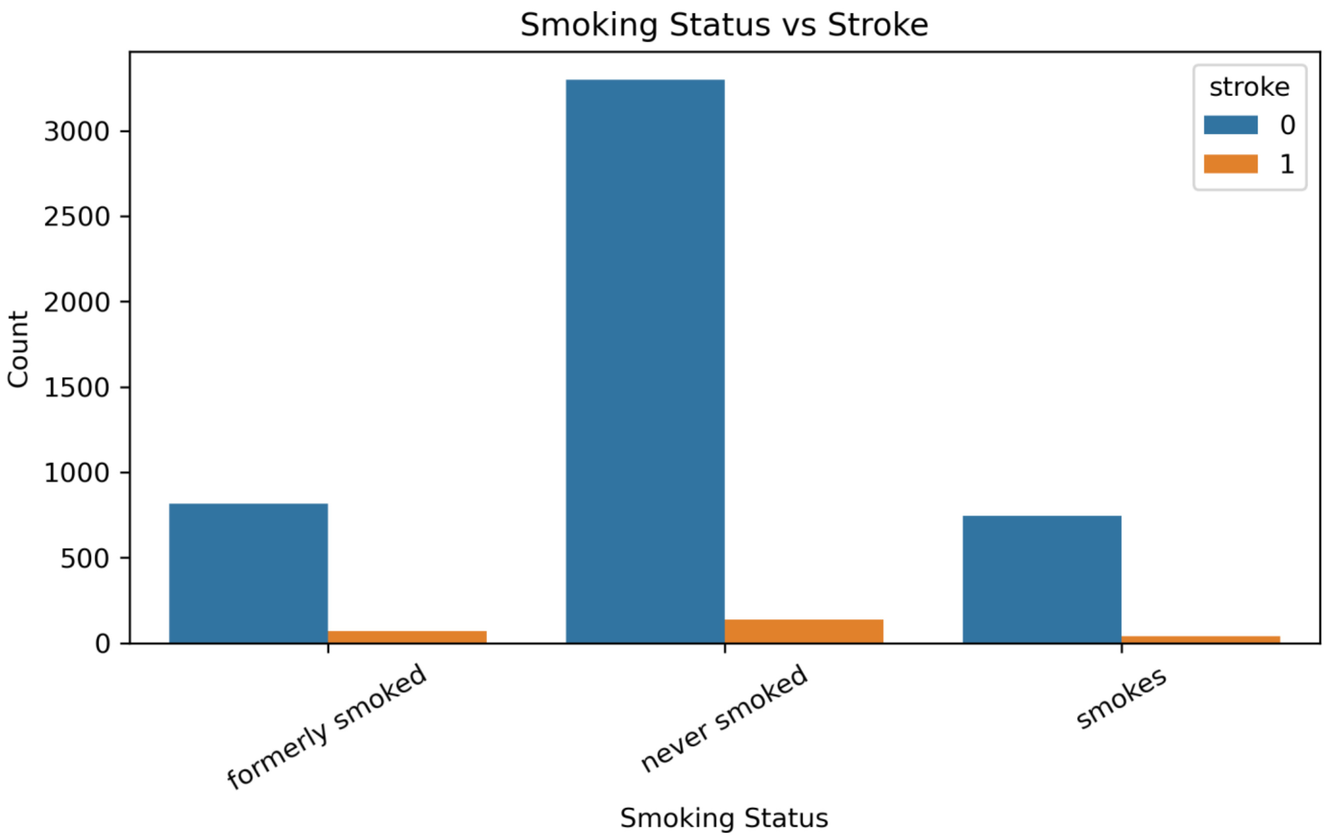
The hypertension versus stroke plot shows that patients with hypertension appear to have a higher proportional risk of stroke compared with patients without hypertension. However, because the dataset is imbalanced, the absolute number of non-stroke cases is still much larger. Therefore, this plot should be interpreted by considering proportions rather than only the height of the bars. Hypertension is still an important clinical feature for stroke prediction.

Figure 6: Heart Disease vs Stroke



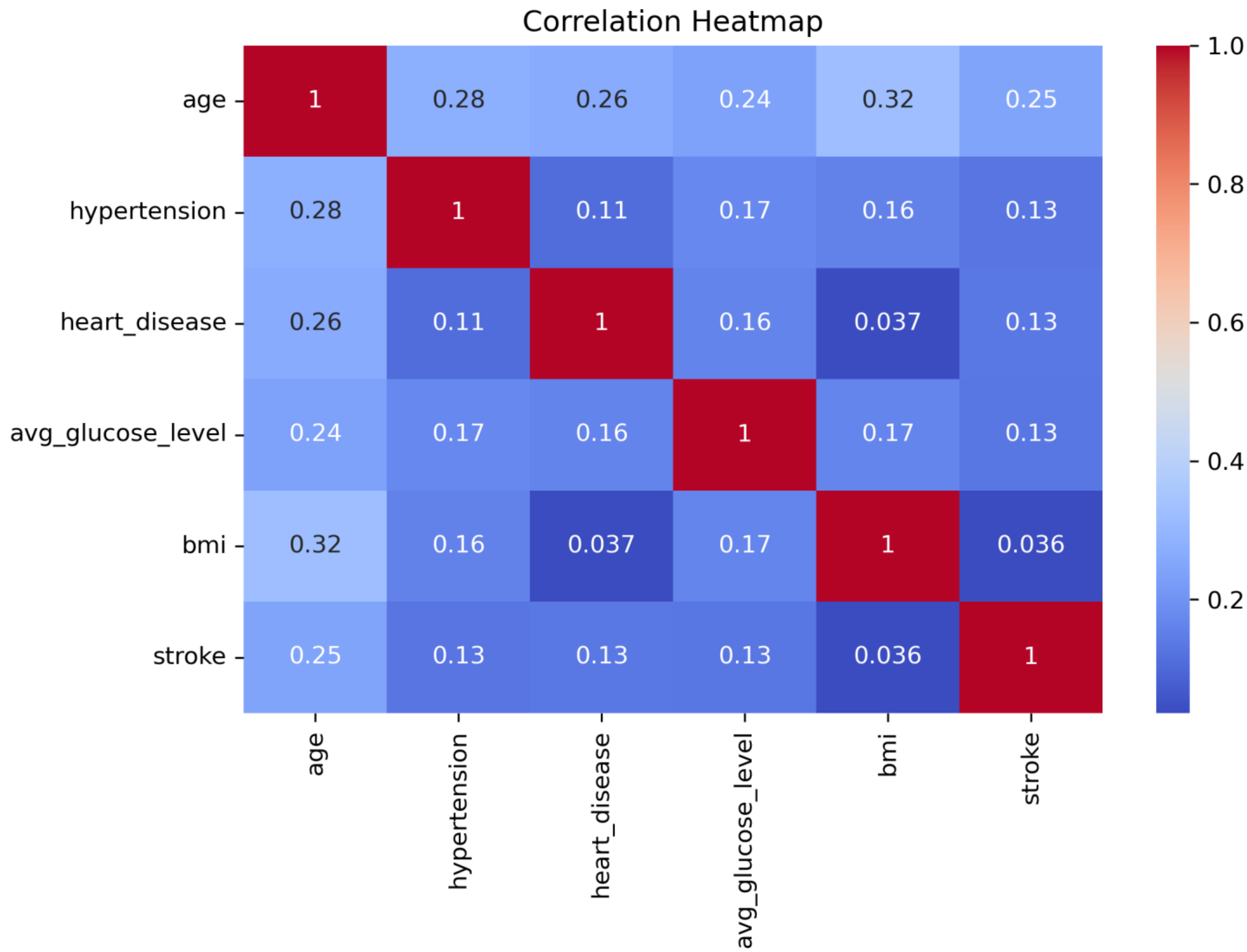
The heart disease versus stroke plot shows that stroke cases are proportionally more common among patients with heart disease than among patients without heart disease. This suggests that heart disease may contribute to stroke risk. Like the hypertension plot, the majority class dominates the counts, so the interpretation should focus on the relationship between heart disease and the stroke proportion rather than total bar height.

Figure 7: Smoking Status vs Stroke



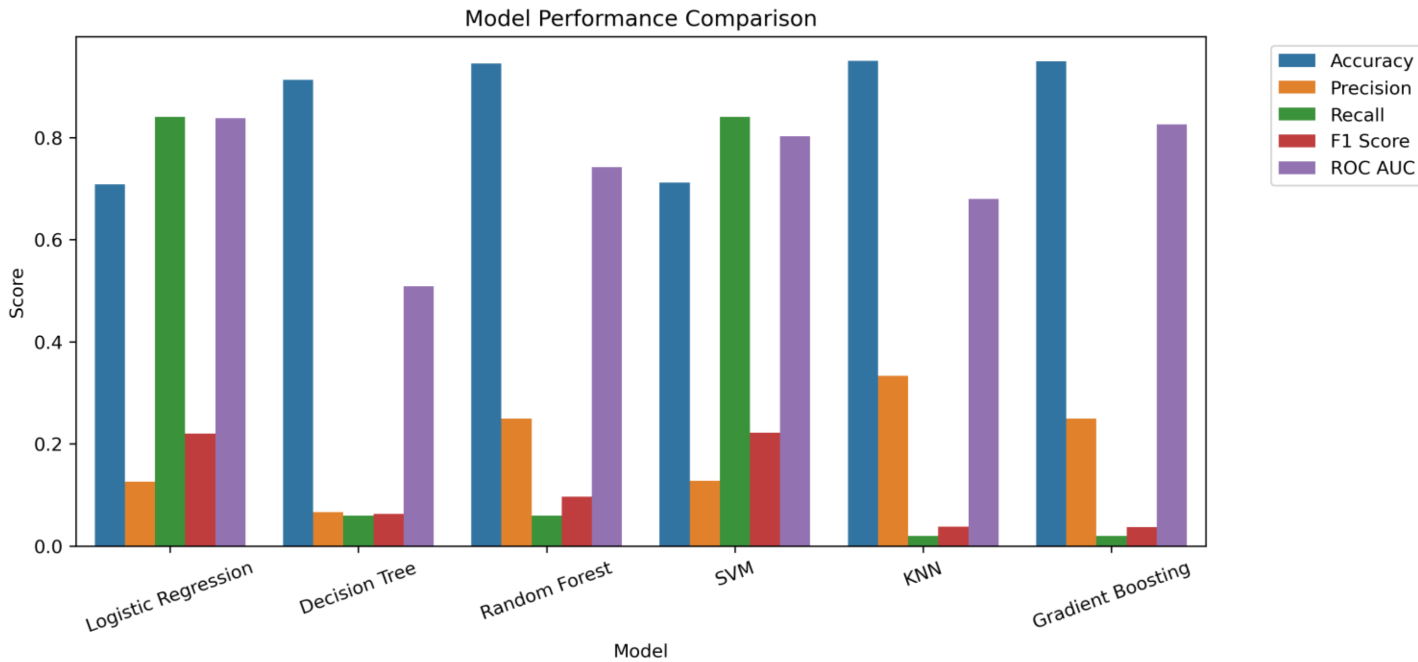
The smoking status plot compares stroke and non-stroke cases across smoking categories. The plot shows that smoking status alone does not clearly separate stroke and non-stroke patients. However, it may still provide useful information when combined with other variables such as age, hypertension, heart disease, BMI, and glucose levels. Therefore, smoking status was kept as part of the model input.

Figure 8: Correlation Heatmap



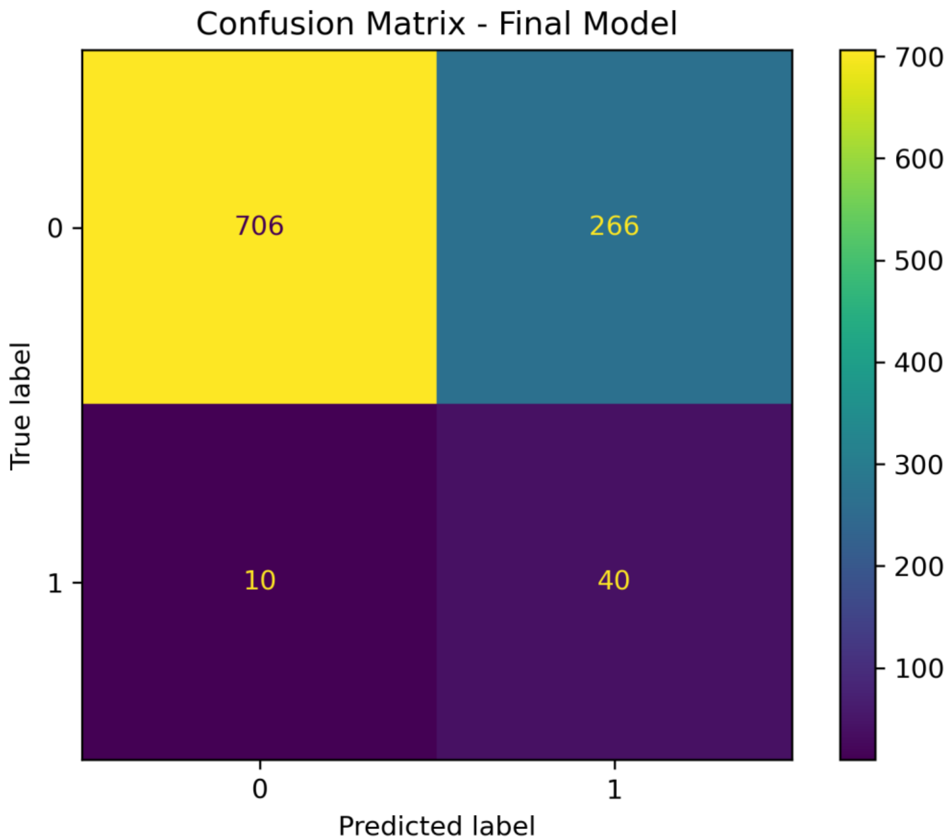
The correlation heatmap shows the relationship between numerical variables. Age has the strongest positive correlation with stroke compared with the other numerical features. Hypertension, heart disease, and average glucose levels also show positive but weaker correlations with stroke. BMI has a much weaker relationship with stroke in this dataset. The heatmap confirms that no single variable fully explains stroke risk, which supports the need for a machine learning model that combines multiple features.

Figure 9: Model Performance Comparison



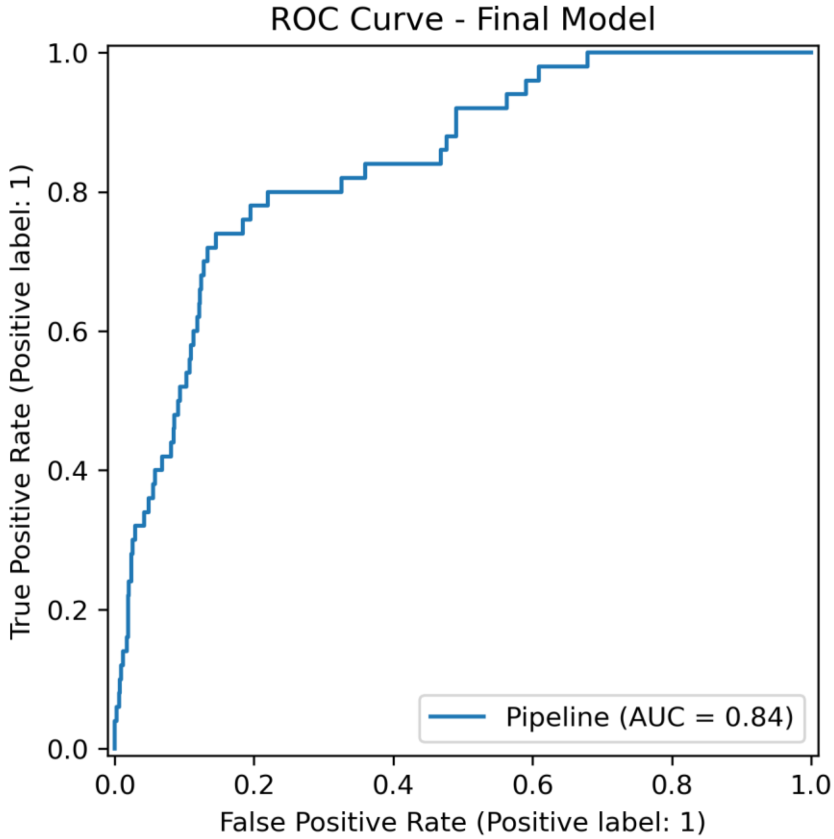
The model performance comparison chart compares the six trained models using accuracy, precision, recall, F1-score, and ROC AUC. Some models, such as KNN, Random Forest, and Gradient Boosting, achieved high accuracy but very low recall for the stroke class. This means they performed well on the majority of non-stroke classes but missed many actual stroke cases. Logistic Regression and SVM achieved better recall, making them more suitable for this medical screening problem. This plot demonstrates why accuracy alone should not be used to select the final model.

Figure 10: Confusion Matrix



The confusion matrix shows the number of correctly and incorrectly classified patients. The model correctly classified 706 non-stroke patients and 40 stroke patients. It missed 10 stroke patients, which means 10 actual stroke cases were predicted as non-stroke. It also incorrectly classified 266 non-stroke patients as stroke-risk. This shows that the model prioritizes detecting stroke cases, even though this increases false positives. For a screening tool, this trade-off can be acceptable because detecting possible stroke-risk patients is more important than minimizing all false alarms.

Figure 11: ROC Curve



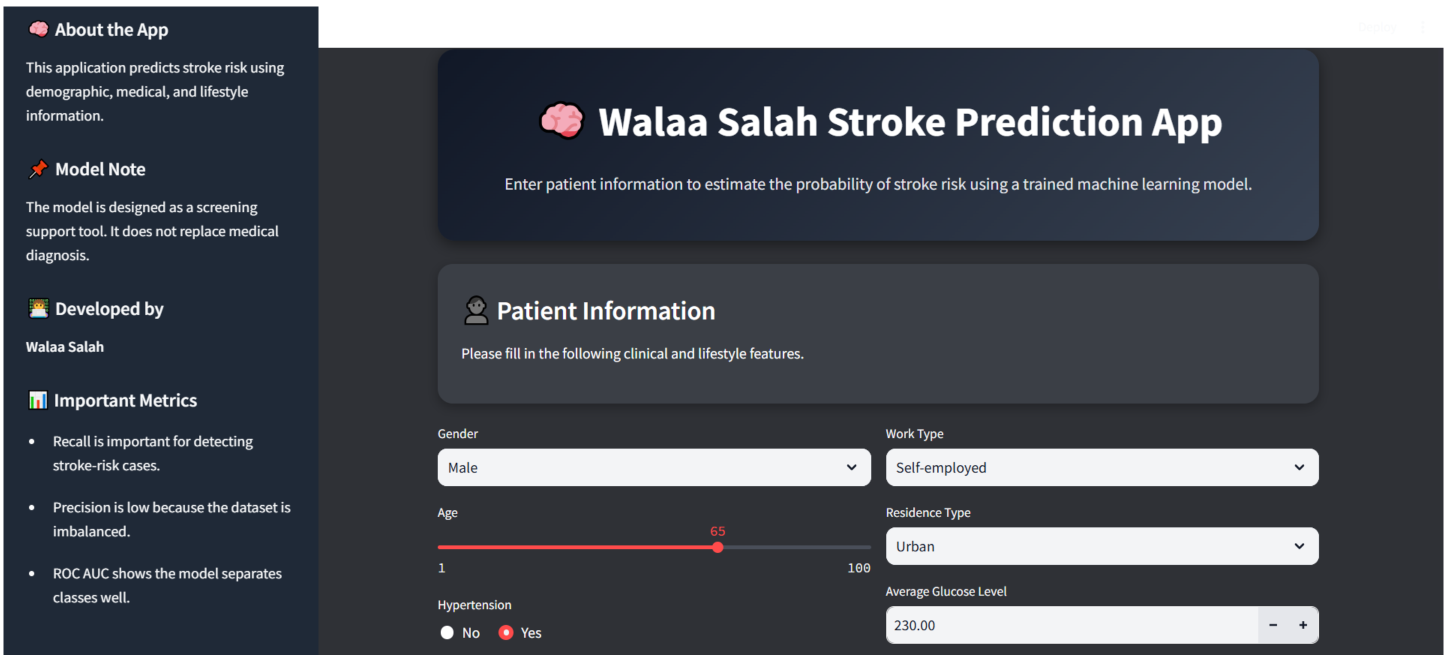
The ROC curve shows the ability of the final model to separate stroke-risk patients from non-stroke patients across different classification thresholds. The model achieved a ROC AUC of approximately 0.84, which indicates good discrimination ability. This means the model is generally able to assign higher risk scores to stroke cases than to non-stroke cases. The ROC curve supports the conclusion that the model has useful predictive ability, even though the dataset is imbalanced.

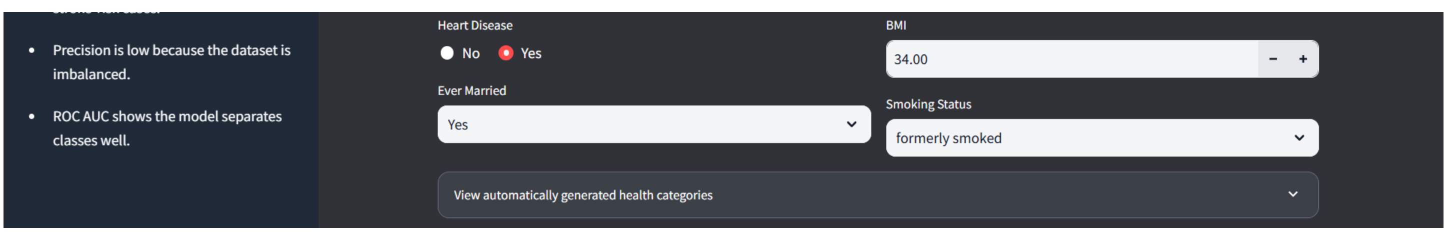
Model Deployment

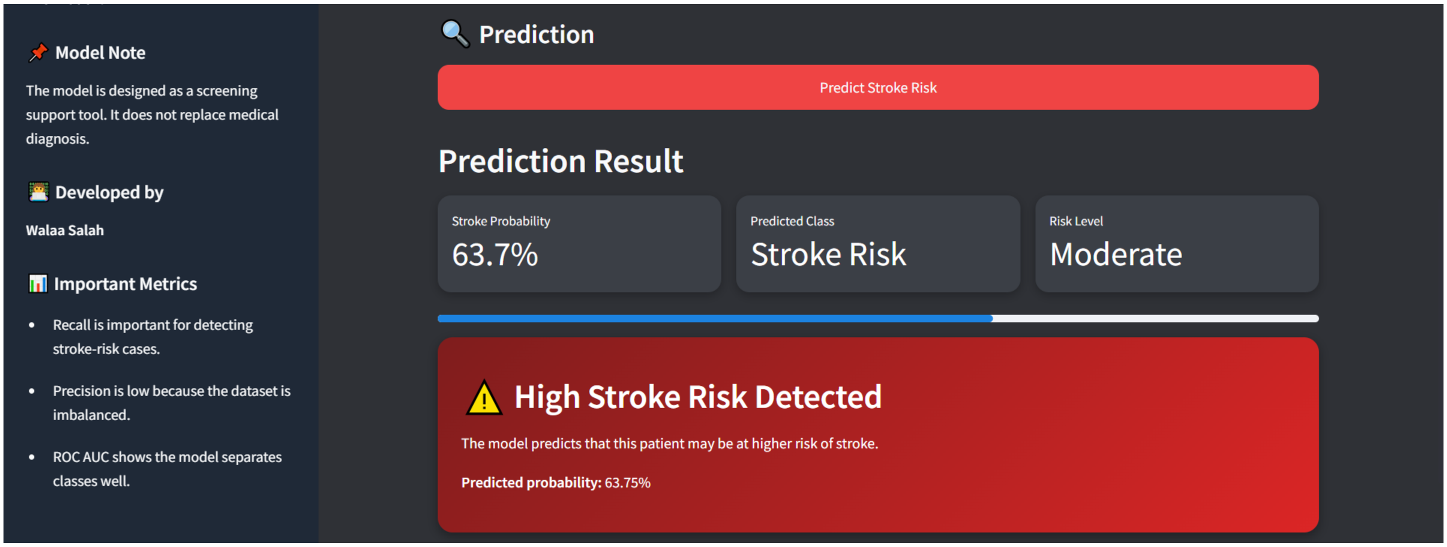
The final model was deployed using Streamlit. A Streamlit application was created to allow users to enter patient information such as gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, BMI, and smoking status. The application also automatically generates engineered features such as age group, BMI category, and glucose risk category.

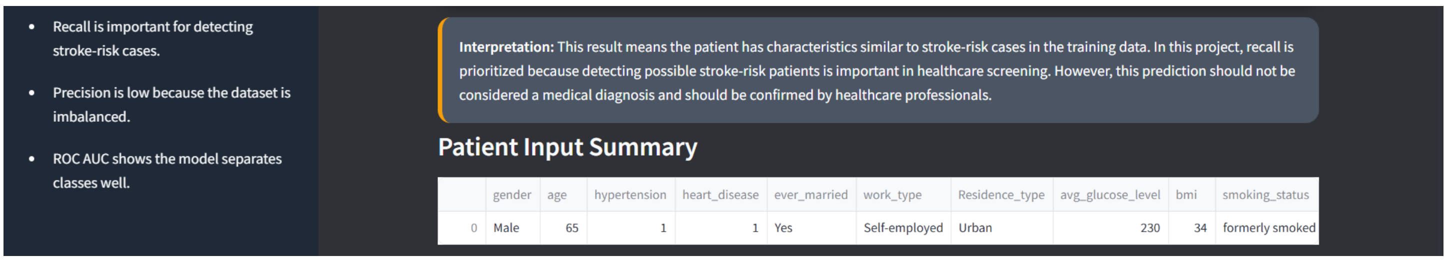
The application loads the trained model from the saved file best_stroke_model.pkl and uses the same preprocessing steps applied during training. After the user enters patient data and clicks the prediction button, the app displays whether the patient is predicted to have high stroke risk or low stroke risk. It also displays the predicted probability of stroke.

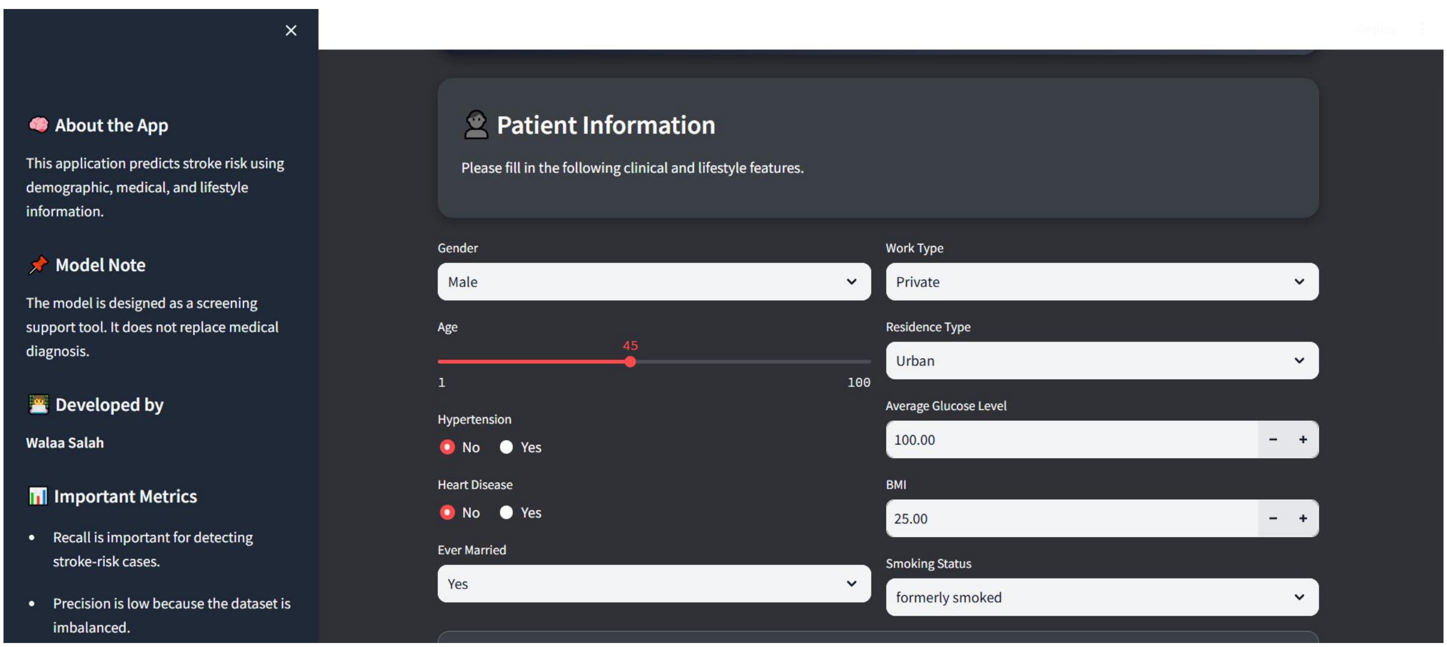
The app was designed to be user-friendly and visually clear, with a dark grey background, a professional title, input sections, risk probability, and interpretation of text. This deployment step makes the machine learning model interactive and easy to demonstrate.

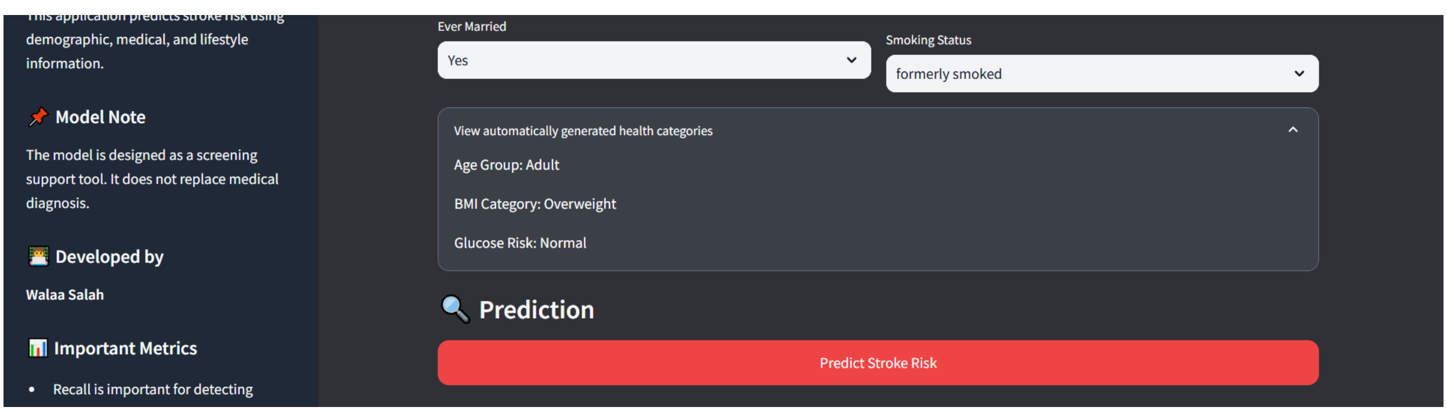


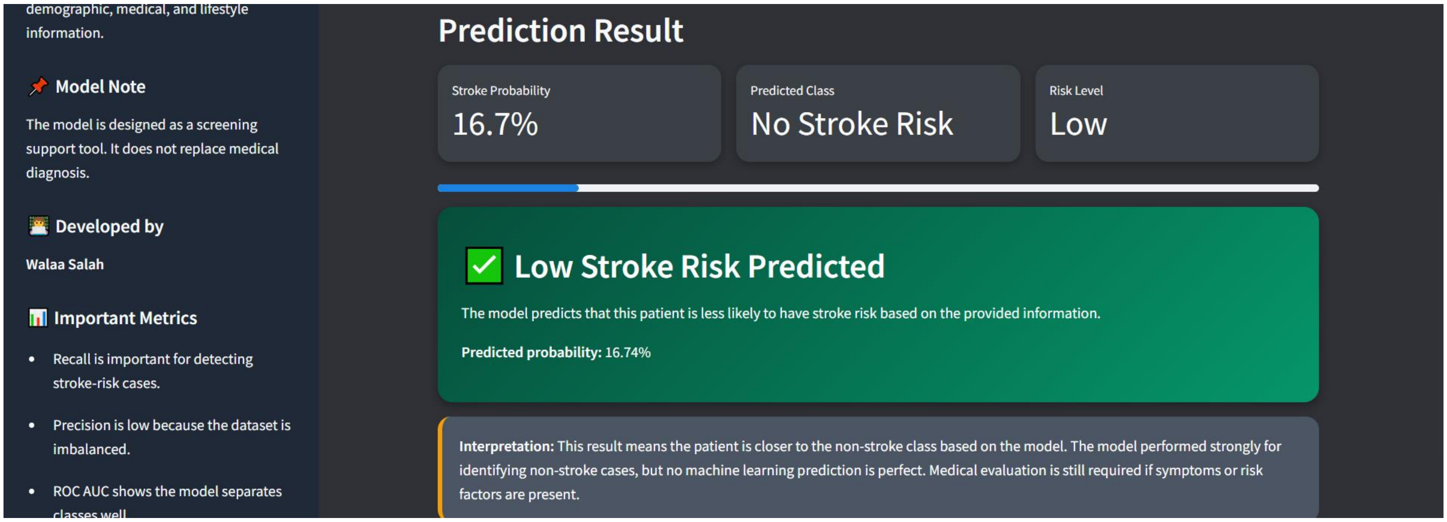



Model Note
The model is designed as a screening support tool. It does not replace medical diagnosis.
Developed by
Walaa Salah
Important Metrics
Recall is important for detecting stroke-risk cases.
Precision is low because the dataset is imbalanced.
ROC AUC shows the model separates classes well.
Prediction
Predict Stroke Risk
Prediction Result
Stroke Probability
63.7%
Predicted Class
Stroke Risk
Risk Level
Moderate
High Stroke Risk Detected
The model predicts that this patient may be at higher risk of stroke.
Predicted probability: 63.75%


Recall is important for detecting stroke-risk cases.
Precision is low because the dataset is imbalanced.
ROC AUC shows the model separates classes well.
Interpretation: This result means the patient has characteristics similar to stroke-risk cases in the training data. In this project, recall is prioritized because detecting possible stroke-risk patients is important in healthcare screening. However, this prediction should not be considered a medical diagnosis and should be confirmed by healthcare professionals.
Patient Input Summary
gender age hypertension heart_disease ever_married work_type Residence_type avg_glucose_level bmi smoking_status
0 Male 65 1 1 Yes Self-employed Urban 230 34 formerly smoked


About the App
This application predicts stroke risk using demographic, medical, and lifestyle information.
Model Note
The model is designed as a screening support tool. It does not replace medical diagnosis.
Developed by
Walaa Salah
Important Metrics
Recall is important for detecting stroke-risk cases.
Precision is low because the dataset is imbalanced.
Patient Information
Please fill in the following clinical and lifestyle features.
Gender
Male
Age
45
1
100
Hypertension
No Yes
Heart Disease
No Yes
Ever Married
Yes
Work Type
Private
Residence Type
Urban
Average Glucose Level
100.00
BMI
25.00
Smoking Status
formerly smoked


demographic, medical, and lifestyle information.
Model Note
The model is designed as a screening support tool. It does not replace medical diagnosis.
Developed by
Walaa Salah
Important Metrics
Recall is important for detecting
Ever Married
Yes
Smoking Status
formerly smoked
View automatically generated health categories
Age Group: Adult
BMI Category: Overweight
Glucose Risk: Normal
Prediction
Predict Stroke Risk


demographic, medical, and lifestyle information.
📌 Model Note
The model is designed as a screening support tool. It does not replace medical diagnosis.
Developed by
Walaa Salah
📊 Important Metrics
Recall is important for detecting stroke-risk cases.
Precision is low because the dataset is imbalanced.
ROC AUC shows the model separates classes well.
Prediction Result
Stroke Probability
16.7%
Predicted Class
No Stroke Risk
Risk Level
Low
✅ Low Stroke Risk Predicted
The model predicts that this patient is less likely to have stroke risk based on the provided information.
Predicted probability: 16.74%
Interpretation: This result means the patient is closer to the non-stroke class based on the model. The model performed strongly for identifying non-stroke cases, but no machine learning prediction is perfect. Medical evaluation is still required if symptoms or risk factors are present.

Conclusion

This project successfully demonstrated a complete machine learning workflow applied to stroke risk prediction. Starting from the raw dataset, the data was cleaned, explored, engineered, preprocessed, and used to train several classification models. The project also included model comparison, hyperparameter tuning, evaluation, visualization, and deployment using Streamlit.

The analysis showed that age, hypertension, heart disease, and average glucose levels are important factors related to stroke risk. The final model achieved good recall and ROC AUC, meaning it was useful for detecting many stroke-risk patients. However, the low precision shows that the model also produces false positives, mainly because the dataset is highly imbalanced. This limitation should be clearly considered when interpreting the results.

Overall, the model is suitable as a screening support tool that can help identify patients who may require further medical evaluation. It should not be considered a replacement for professional medical diagnosis. Future improvements could include using a larger and more balanced dataset, applying advanced imbalance-handling techniques such as SMOTE, testing more algorithms, improving feature engineering, and deploying the application publicly using Streamlit Community Cloud.